Student Absenteeism

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Description

Students that are chronically absent miss direct instruction, participation in classroom activities, and engagement with peers (Ginsburg, Jordan, & Chang, 2014), score lower on math and reading assessments (Gottfried, 2019), experience increased class failure rates (Allensworth & Easton, 2007), require additional support or interventions to catch up with missed lessons and material (Balkis, Arslan, & Duru, 2016), risk falling behind, disengaging from school, and potentially dropping out (Singer, Pogodzinski, Lenhoff, & Cook, 2021). The impacts and potential costs of absenteeism are apparent. This model provides a valuation of the per student opportunity cost of absence at public school district level.

Where/When to Apply the Method

Two conditions – where and when - are required to use this data. The first condition of use, projects where student absence is part of the scope. The second condition of use, when data assumptions on durations absenteeism and number of students experiencing absenteeism are available. Having these conditions met allows for project researchers to multiply the duration of absences by the cost per durations (hours or days).

Appropriate Ecosystem Services/Disservices

- Water/Air/Soil Quality – Air quality, snow fall, and heat waves are measurable. Depending on the quality of data collection (performed by a CBO preferably), EE can create generalizable assumptions about student absences in a community during weather events. Said assumptions would be backed by literature showing a connection between weather events and student absenteeism/academic performance.

Note that student absenteeism likely only reflects the partial impact of environmental impacts. For instance, in cases of natural disaster or mold/pollution induced asthma that may cause students to miss school, it an opportunity cost. Also, seasonality may determine whether absenteeism is relevant (e.g., disasters during summer vacations would not necessarily lead to lost school days).

Data Requirements

Tabular Data:

- (https://www.census.gov/data/tables/2019/econ/school-finances/secondary-education-finance.html), the *2019 Public Elementary-Secondary Education Finance Data* provides total annual expenditure and enrolled students for each public school district. Duration of student absenteeism
- (https://nces.ed.gov/surveys/sass/tables/sass0708_035_s1s.asp) The National Center for Education Statistics (NCES) and their Schools and Staffing Survey (SASS) for the 2007-2008 school year, provides data on average number of hours in the school day and average number of days in the school year for public schools, by state: 2007–08.

Methodology

This method helps calculate the cost of students being absent from school by considering the time they miss. To use this method effectively, it's best to gather information on how long students are absent and how many of them are affected. However, it's important to note that this method alone cannot give a

Commented [AF1]: Then, how are the $/student/day values used? Is it applied to number of missed school days due to disasters/mold/whatever? Where does one get that data? Essentially, how do you get to a total $/year value from this?

complete picture of the cost of student absenteeism for a specific project. It only calculates the cost per student on a daily or hourly basis. To get the full cost, additional data and calculations are needed to understand the true impact of student absenteeism in a specific project.

Data on student absenteeism opportunity costs is not publicly available. Using data from SASS This method calculates the broad daily student absentee opportunity cost at a school district level for every public school district in The United States. My approach rescales the annual budgets of public school from $/district/year to $/student/day. It should be noted that this approach valuates the broad opportunity cost for student missing school. Rescaling to the daily opportunity cost requires a series of intermediate steps.

The first step is to collect data on the total annual expenditure and enrolled students in each public school district. This is a rather simple process. Heading Census.gov (https://www.census.gov/data/tables/2019/econ/school-finances/secondary-education-finance.html), the *2019 Public Elementary-Secondary Education Finance Data* provides total annual expenditure and enrolled students for each public school district. Collection of data is coordinated by the National Center for Education Statistics (NCES), which is part of the U.S. Department of Education. Below is a small sample of the data collected from the *2019 Public Elementary-Secondary Education Finance Data report.*

With school district total annual expenditures and total number of enrolled students known, initial calculations can begin. Borrowing the method detailed on page 247 of the OECD Education at a Glance 2022: OECD Indicators. "Expenditure per student on educational institutions at a particular level of education is calculated by dividing total expenditure on educational institutions at that level by the corresponding full-time equivalent enrolment. Only educational institutions and programs for which both enrolment and expenditure data are available are taken into account" (OECD, 2022, p. 247). Thus, the annual school district expenditure per student ($/year/student) has been calculated.

Table 1: Public School Expenditure Sample Spreadsheet

State_Name	District_NAME	Enrolled Students (v33)	Annual District Expenditure (TotalEXP)	Annual District Expenditure per Student (EP)
Alabama	AUTAUGA COUNTY SCHOOL DISTRICT	9094	$83,714.00	$9,205.41
Alabama	BALDWIN COUNTY SCHOOL DISTRICT	32267	$417,727.00	$12,945.95
Alabama	BARBOUR COUNTY SCHOOL DISTRICT	755	$9,606.00	$12,723.18
Alabama	EUFAULA CITY SCHOOL DISTRICT	5427	$39,926.00	$7,356.92

Note. This table shows Public School District Expenditure data provided by National Center for Education Statistics (NCES). Additionally, column "Annual District Expenditure per Student (EP)" is a calculated field.
$\frac{TotalEXP}{v33} = EP$

The next step is to scale down from an annual school district expenditure per student to a daily school district expenditure per student. Before the daily school district expenditure per student ($/day/student) can be calculated, it must be determined how many annual and daily hours a student spends in class. According to data from the National Center for Education Statistics (NCES) and their Schools and Staffing Survey (SASS) for the 2007-2008 school year, available at https://nces.ed.gov/surveys/sass/tables/sass0708_035_s1s.asp, Table 2 provides information on average number of hours in the school day and average number of days in the school year for public schools, by state: 2007–08. By multiplying the state class hours per day by the average state class days per year, the average state class hours per year is derived. The benefit of using NCES state class statistics is the ability to control for discrepancies in how states govern their education systems.

Table 2: Average number of hours in the school day and average number of days in the school year for public schools, by state: 2007–08

State	Hours /Day (HD)	Days /Year (DY)	Hours /Year (HY)
USA	6.66	178.77	1190.18
AL	7.03	180.00	1265.40
AK	6.48	180.00	1166.40
AZ	6.43	181.00	1163.83
AR	6.89	179.00	1233.31
CA	6.24	181.00	1129.44
CO	7.01	171.00	1198.71
CT	6.47	181.00	1171.07
DE	6.68	181.00	1209.08
DC	6.91	181.00	1250.71
FL	6.43	184.00	1183.12
GA	6.79	181.00	1228.99
HI	6.26	179.00	1120.54
ID	6.63	173.00	1146.99
IL	6.50	177.00	1150.50
IN	6.77	180.00	1218.60
IA	6.85	180.00	1233.00
KS	6.98	178.00	1242.44
KY	6.69	180.00	1204.20
LA	7.08	178.00	1260.24
ME	6.47	176.00	1138.72
MD	6.59	180.00	1186.20
MA	6.45	180.00	1161.00
MI	6.56	178.00	1167.68
MN	6.28	176.00	1105.28
MS	6.99	181.00	1265.19
MO	6.70	177.00	1185.90
MT	6.79	179.00	1215.41
NE	6.92	178.00	1231.76
NV	6.30	180.00	1134.00
NH	6.54	180.00	1177.20
NJ	6.44	181.00	1165.64
NM	6.85	177.00	1212.45
NY	6.59	182.00	1199.38
NC	6.75	180.00	1215.00
ND	6.58	176.00	1158.08
OH	6.61	180.00	1189.80
OK	6.63	176.00	1166.88
OR	6.57	172.00	1130.04
PA	6.43	181.00	1163.83
RI	6.27	180.00	1128.60
SC	6.92	181.00	1252.52
SD	6.83	173.00	1181.59
TN	7.03	180.00	1265.40
TX	7.17	180.00	1290.60
UT	6.28	182.00	1142.96
VT	6.66	177.00	1178.82
VA	6.62	181.00	1198.22
WA	6.22	179.00	1113.38
WV	6.87	182.00	1250.34
WI	6.91	180.00	1243.80
WY	6.86	175.00	1200.50

Note. Table 2 shows the rate of time by which students spend in class. Rate of student class time is denominated in hours per day (HD), days per year (DY), and hours per year (HY). hours per year (HY) is a calculated field. $\boldsymbol{HD} \times \boldsymbol{DY} = \boldsymbol{HY}$

Thus far, the district Annual Expenditure per Student and state Annual Hours in Class have been calculated. Below, figure 1 uses the Alabama, Autauga County School District to visualize how the two calculated fields mentioned are created.

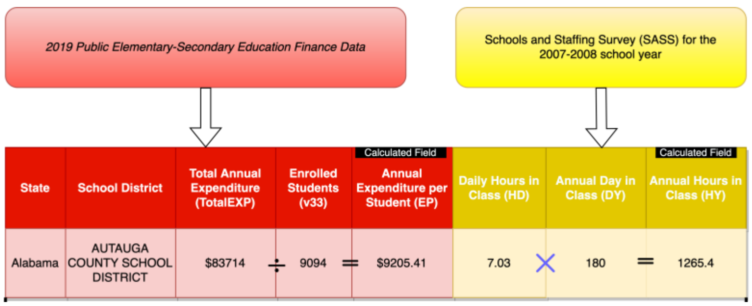

Figure 1: Sourcing and Calculations of Education Data

Next, the "annual expenditure per student (EP)" is divided by the "annual class hours per year (HY)". This derives the "Hourly Expenditure per Student" ratio. Finally, the state average class hours are multiplied by the hourly expenditure per student. Below, this step can be seen in figure 2.

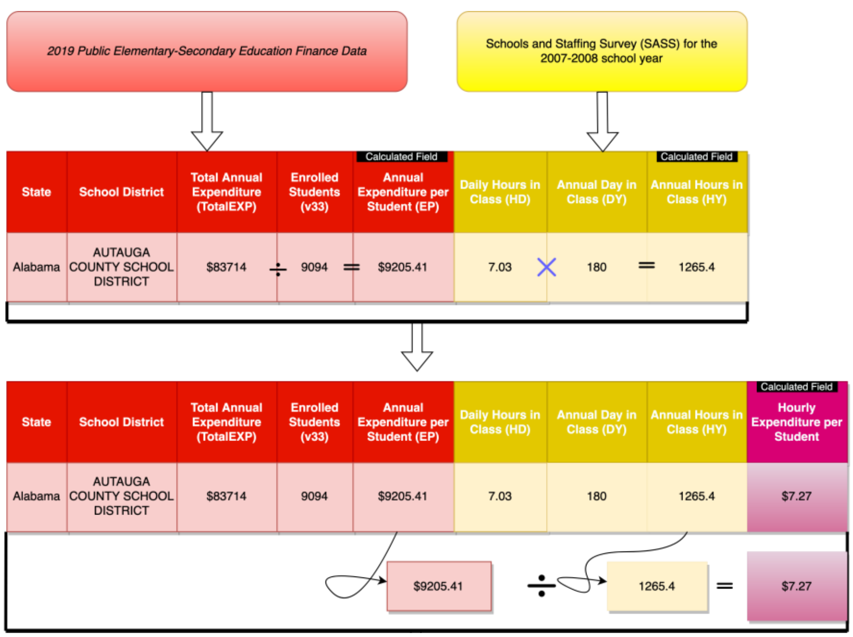

Figure 2: Hourly Expenditure per Student Ratio Calculation

The final step is to calculate the "Daily Expenditure per Student" ratio. This last calculation multiplies the "Daily Hours in Class (HD)" by the "Annual Expenditure per Student". This produces the "Daily Expenditure per Student" ratio. This process can be seen below in figure 3.

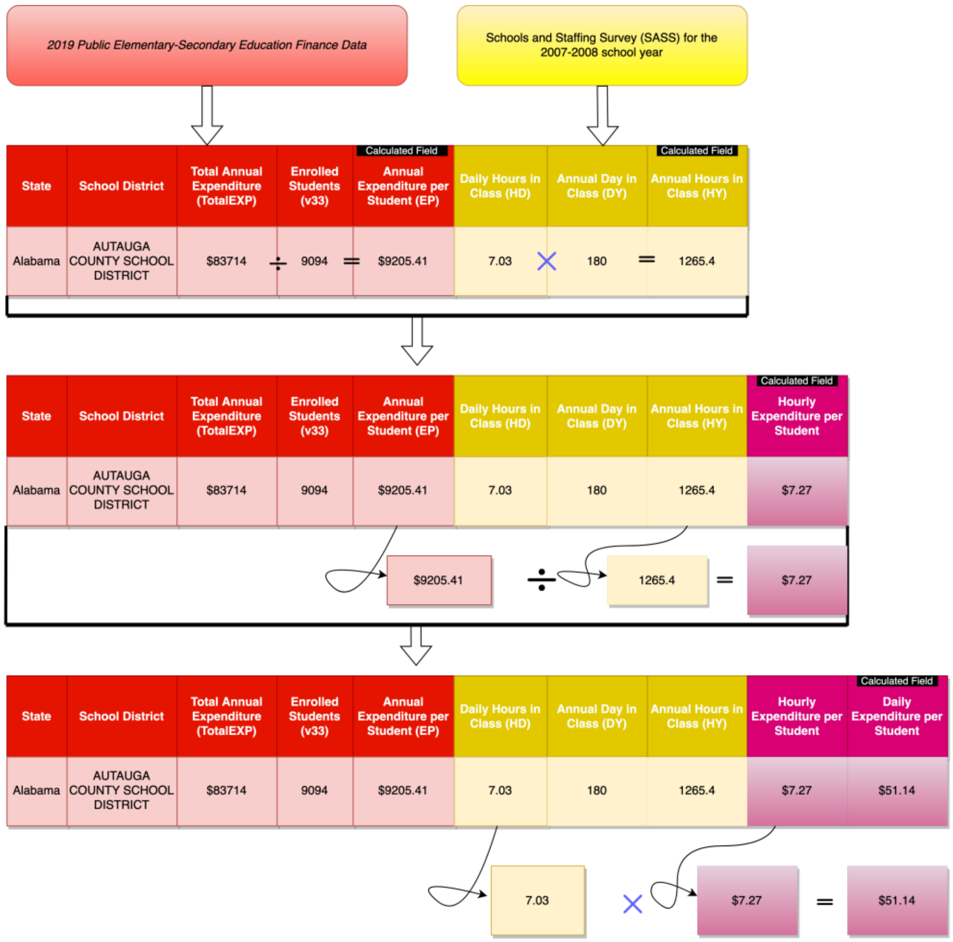

Figure 3: Daily Expenditure per Student Ratio Calculation

In total, this methodology walks through the process of obtaining the data through links provides, calculating the obtained data to derive the daily expenditure per student. The results of this methodology will be of use in calculating the daily opportunity cost of absenteeism. The results of this method shall serve as the base of future methods that seek to quantify project specific absenteeism costs.

Limitations

1. **Assumptions and simplifications:** The method assumes that the *2008 Schools and Staffing Survey* accurately reflects the present-day instructional hour requirements of each state. Further, this method uses 2019 – pre pandemic data – to calculate the student absenteeism cost.

Supporting Literature and Resources

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